Akoizo Original Practice - Form 3 - Answer key

Division B | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Parallel resistors share…

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. Voltage

B. Always current

C. Always power

D. Always resistance

Correct: A

2. (2 points) KCL expresses…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. Charge conservation

B. Equal current in all branches

C. Energy creation

D. Resistance addition only

Correct: A

3. (2 points) Which change best tests the cause of the dark LED?

Unit: Fault finding with switches, fuses, and diodes.
Worked-case data: Supply: The battery reads 6 V. Branch A: Its LED lights through a resistor. Branch B: Its LED is reversed and remains dark.
Separate investigation observations:
Supply: The battery reads 6 V.
Branch A: Its LED lights through a resistor.
Branch B: Its LED is reversed and remains dark.

A. Remove both current-limiting resistors

B. Reverse the LED with power disconnected, retaining its resistor

C. Short the battery terminals

D. Replace the working battery without checking wiring

Correct: B

4. (2 points) Two 100 Ω resistors in series give…

A. 200 Ω

B. 50 Ω

C. 100 Ω

D. 10000 Ω

Correct: A

5. (2 points) Negative solved current means…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. Opposite the assumed direction

B. Impossible circuit necessarily

C. Negative charge disappears

D. No voltage exists

Correct: A

6. (4 points) Explain the evidence supporting your answer, including the calculation or comparison.

Unit: Fault finding with switches, fuses, and diodes.
Worked-case data: Supply: The battery reads 6 V. Branch A: Its LED lights through a resistor. Branch B: Its LED is reversed and remains dark.
Separate investigation observations:
Supply: The battery reads 6 V.
Branch A: Its LED lights through a resistor.
Branch B: Its LED is reversed and remains dark.

answer: The working branch shows that the supply can power an LED. Compare LED polarity and connections in the two branches. Disconnect power, correct the reversed LED, and retest with its resistor. (4 points).

7. (2 points) An open switch can have…

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. Voltage with zero current

B. Infinite real power

C. Always zero voltage

D. Consumed charge

Correct: A

8. (2 points) Resistor dissipation is…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. I^2R

B. Always negative

C. Independent of current

D. Equal to charge

Correct: A

9. (4 points) Explain "Open circuit" in your own words.

Unit: Fault finding with switches, fuses, and diodes.
Worked-case data: Supply: The battery reads 6 V. Branch A: Its LED lights through a resistor. Branch B: Its LED is reversed and remains dark.
Separate investigation observations:
Supply: The battery reads 6 V.
Branch A: Its LED lights through a resistor.
Branch B: Its LED is reversed and remains dark.

answer: A broken conducting path. (4 points).

10. (4 points) Why trace nodes first?

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

answer: Topology follows connections, not visual placement. (4 points).

11. (4 points) Write a source-two-resistor loop equation.

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

answer: Vs - IR_1 - IR_2 = 0 for traversal with current. (4 points).

12. (6 points) Solve this case and explain your reasoning: Supply: The battery reads 6 V. Branch A: Its LED lights through a resistor. Branch B: Its LED is reversed and remains dark.

reasoning: The working branch shows that the supply can power an LED.
Compare LED polarity and connections in the two branches.
Disconnect power, correct the reversed LED, and retest with its resistor. (3 points).

conclusion: Reverse the LED with power disconnected, retaining its resistor (3 points).

13. (4 points) Find 100 Ω parallel 100 Ω.

answer: 50 Ω. (4 points).

14. (4 points) Why avoid dependent equations?

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

answer: They repeat information and do not determine additional unknowns. (4 points).

15. (4 points) Name a useful reference feature.

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

answer: An equation beside its units, assumptions, and a checked example. (4 points).

16. (4 points) What does a relay separate?

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

answer: Control and contact/load functions, subject to its actual isolation and ratings. (4 points).

17. (4 points) Why isolate an unknown for resistance measurement?

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

answer: Other paths can alter the reading. (4 points).

18. (4 points) Does zero field imply zero potential?

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

answer: No; vector and scalar sums can differ. (4 points).